

ND I MECHANICAL ENGINEERING : MEC 106 (ENGINEERING METALLURGY & MATERIALS)

1.1 DEFINITION OF SOME USEFUL TERMINOLOGIES:

- a) *Engineering*: is the application of science by using your initiative and creativity to design, plan, construct and maintains building; structure, machines and other manufactured goods.
- b) *Metallurgy*: is the study of chemical reaction that takes place in the production of metal and its alloys, such as physical, chemical and mechanical behavior of metal
- c) *Science*: is the study of the physical and natural world and phenomena, especially by using systematic observation and experiment.
- d) *Metallurgical Engineering*: is the branch of engineering that deals with the study of the structure and properties of metal, their extraction from the ground and the procedures of refining, alloying and making things from them.
- e) *Material Science*: it is the study of characteristics of nature, behavior and uses of various materials such as plastics and metals as applied to science and technology.
- f) *Material Engineering*: is the branch of engineering that deals with the study of processing materials such glass, plastics and metals to meet the demand of the users.
- g) *Mechanical Engineering*: is the branch of engineering that deals with design, production and use of machinery and tools as well as the generation and transmission of heat and mechanical power.
- h) *Mechanical Metallurgy*: is the study of science and technology of metal behaviors in relation to mechanical force impose on them, this include rolling, extrusion, bending and deep drawing.
- i) *Production Metallurgy*: is the planning and controlling of mechanical means of changing the shape, condition and relationship of material towards greater effectiveness and value

j) *Extractive Metallurgy*: this is the branch of metallurgy that deals with the survey being carried out on minerals ores prior to its excavation or mining be either surface or deep mining methods.

k) *Chemical Metallurgy*: the branch of metallurgy that deals with reaction taking within the context of minerals ores while various chemical are being added aspect leads to the extraction of pure minerals from its ores.

2.1

METALS

A metal is defined as an element with valences of +1, +2, and +3. However can be best defined by the nature of the bond between the atoms that make up the crystal. It can also be defined as a solid that comprised of an atom held together in a matrix of electrons, the electrons associated with each individual atom and are free to move through the volume of the crystal or piece of metal. This is why metals are good conductors of electricity; current flow requires a flow of electrons

2.1.1

Physical Properties of Metals

Physical properties of elements are dependent on the following factors

- The arrangement of their atoms or molecules in crystal lattice when in solid state
- The bonds that bind the atom or molecule in solid, liquid and gaseous states

Metals are solid at room temperature and exist as crystal lattices in which the atoms are held together by strong metallic bonds, metals have following physical and mechanical properties:

- High boiling and melting point (Physical)
- Conduct heat and electricity (Physical)
- Ductile; that is easily drawn into wires (Mechanical)
- Possess high tensile strength. (Mechanical)
- Malleable; that is, they are easily hammered into sheets (Mechanical)
- Have characteristics luster (Physical)
- Hard but not brittle (easily broken) with great tensile strength (Mechanical)

VIII. Relative high density (Physical)

ts 257°C

However, some metals do not exhibit all these properties for example, mercury is a liquid with a freezing point of -39 °C, and sodium and potassium are light, soft, metals with low melting point of 97 °C and 63 °C respectively. Certain elements like Boron, Silicon and Germanium possess some properties falling between metals and Non-metals, which is referred to as METALLOIDS. Many metals are used in the "Pure" state for electroplating, durable goods, tools and electrical works or devices which include Cr, Ni, Cd, Sn, Zn, Os, Re and Rh.

Among the non-metals, carbon has an exceptional characteristics as it is extremely hard with a very high melting point when it exists as diamond and it is a good conductor of electricity when it exists as graphite. Carbon is the brain of organic chemistry called hydrocarbon. Carbon is used in industrial application for motor brush and wears parts. Inert gases and other elements of non-metals are in the elemental form for industrial application for prospective atmosphere and other.

2.1.2

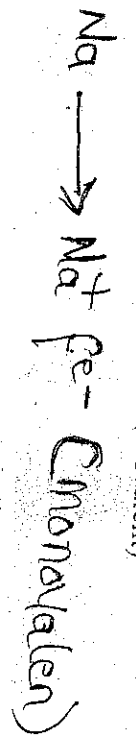
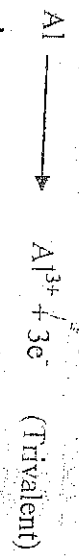
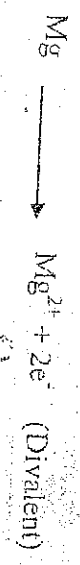
Chemical Properties of Metal

The following are chemical properties of metal, which include:

- I. It acts as a reducing agent (Hydrogen and Carbon)
- II. It reacts with acids to liberate hydrogen and salts, for instance,



- III. Metals ionize very easily by losing its outmost electrons



2.1.3

Alloys

It is a combination of two or more metals (elements). Commercially pure metals find many valuable use e.g. Copper for conductor. Aluminum for foil and Chromium for plate surface. Alloy is a substance that has a mixture of two or more metals or mixture of a metal with a non-metallic material. Alloys steel containing elements other than carbon and above 2 % molybdenum. Alloy substance composed of two or more metals. Alloys, like pure metals, possess metallic luster and conduct heat and electricity well, although not generally as well as do the pure metals of which they are formed. Compounds that contain both a metal or metals and certain non-metals, particularly those containing carbon, are also called alloys. The most important of these is steel. Simple carbon steels consist of about 0.5 percent manganese and up to 0.8 percent carbon, with the remaining material being iron.

An alloy may consist of inter-metallic compound, a solid solution, an intimate mixture of minute crystals of the constituent metallic elements, or any combination of solutions or mixtures of the foregoing. Inter metallic compounds, such as NaAu_2 , CuSn , and Cu_3Al_2 , do not follow the ordinary rules of valence. They are generally hard and brittle; although they have not been important in the past where strength is required: many new developments have made such compounds increasingly important. Alloys consisting of solutions or mixtures of two metals generally have lower melting points than do the pure constituents. A mixture with a melting point lower than that of any other mixture of the same constituents is called a eutectic. The eutectoid, the solid-phase analog of the eutectic, frequently has better physical characteristics than do alloys of different proportions.

The properties of alloys are frequently far different from those of their constituent elements, and such properties as strength and corrosion resistance may be considerably greater for an alloy than for any of the separate metals. For this reason, alloys are more generally used than pure metals. Steel is stronger and harder than wrought iron, which is approximately pure iron, and is used in far greater quantities. The alloy steels, mixtures of steel with such metals as chromium, manganese, molybdenum, nickel, tungsten, and

vanadium, are stronger and harder than steel itself, and many of them are also more corrosion resistant than iron or steel. An alloy can often be made to match a predetermined set of characteristics. An important case in which particular characteristics are necessary is the design of rockets, spacecraft, and supersonic aircraft. The materials used in these vehicles and their engines must be light in weight, very strong, and able to sustain very high temperatures. To withstand these high temperatures and reduce the overall weight, lightweight, high-strength alloys of aluminum, beryllium, and titanium have been developed. To resist the heat generated during reentry into the atmosphere of the earth, alloys containing heat-resistant metals such as tantalum, niobium, tungsten, cobalt, and nickel are being used in space vehicles.

* A wide variety of special alloys containing metals such as beryllium, boron, niobium, hafnium, and zirconium, which have particular nuclear absorption characteristics, are used in nuclear reactors. Niobium-tin alloys are used as superconductors at extremely low temperatures. Special copper, nickel, and titanium alloys, designed to resist the corrosive effects of boiling salt water, are used in desalination plants.

3.1

CHEMICAL BONDS

Bond is a fundamental attractive force that binds atoms and ions in a molecule. It is a property when two or more atoms and ions have a strong attraction for each other. There are several types of chemical bond which include:

- I. Electrovalent or ionic bond
- II. Covalent bond
- III. Metallic bond
- IV. Hydrogen bond

3.1.1 Electrovalent bond

This results when a strong electropositive metal i.e. Na (Sodium) completely transfer its most loosely bounded electron to highly electronegative element i.e. Cl (Chlorine). In solid sodium chloride, the sodium and chloride ions attract each other, for

What is ionic bond. The electrostatic force, the attraction of unlike charges or the repulsion of like charges operates in all direction. Ionic compound tends to have high melting and boiling points because the electrostatic forces that hold the ions in place are fairly strong and extended in all direction.

3.1.2 Covalent bond

This is formed as result of electron sharing or electron pair equally between non-metals. Example is chlorine, fluorine gas molecules, hydrochloride and hydrogen molecules.

3.1.3 Metallic bond: This is bond formed between two metals which constitute an alloy. It is chemical bonds characteristic of metals, in which electrons are shared between atoms and move about in the crystal.

3.1.4

Hydrogen bond: An electrostatic interaction between molecules of compound in which hydrogen atoms are bound to electronegative atoms such as oxygen or nitrogen, the attraction between water molecules due to hydrogen bonds accounts for the relatively high boiling point of water

4.

CORROSION AND ITS PREVENTION

4.1.1. Corrosion

Corrosion is defined as a process whereby metal involves in the heterogeneous chemical or electrochemical surface reaction with environmental and the tendency to return to ionic states usually of lower energy. Corrosion is partial or complete wearing away, dissolving, or softening of any substance by chemical or electrochemical reaction with its environment. The term *corrosion* specifically applies to the gradual action of natural agents, such as air or salt water, on metals.

The most familiar example of corrosion is the rusting of iron, a complex chemical reaction in which the iron combines with both oxygen and water to form hydrated iron oxide. The oxide

is a solid that retains the same general form as the metal from which it is formed and somewhat bulkier, is relatively weak and brittle.

~~Three~~ Three methods may be used to prevent the rusting of iron:

- i. Alloying the iron so that it will be chemically resistant to corrosion;
- ii. Coating it with a material that will react with the corroding substance readily than the iron does and thus, while being consumed, protect the iron;
- iii. Covering it with an impermeable surface coating so that air and water cannot reach it. The alloying method is the most satisfactory but the most expensive example is stainless steel, in which chromium or chromium and nickel with the iron; this alloy is not only absolutely rustproof but will even resist the action of such corrosive chemicals as hot, concentrated nitric acid. The most common example of this method is galvanizing, in which iron is coated with zinc. In the presence of corrosive solutions, an electric potential is set up between the iron and the zinc, causing the zinc to dissolve but protecting the iron as long as any zinc remains. The third method, protection by coating with an impermeable layer, is the least expensive and therefore the most satisfactory as long as no crack appears in the coating. Once the coating cracks, however, rusting proceeds at least as fast as it would have with no protection. If the protective layer is an inactive metal, such as tin or chromium, the electric potential is set up, protecting the layer but acting on the iron and causing the rusting to proceed at an accelerated rate. The most satisfactory coatings are baked enamels; the least expensive are such paints as red lead.

Some metals, such as aluminum, although very active chemically, appear not to corrode under normal atmospheric conditions. Actually, aluminum corrodes rapidly, and a thin, transparent layer of oxide forms on the surface of the metal, protecting it from further corrosion. Lead and zinc, although less active than aluminum, are protected by similar films. Copper, a comparatively inactive metal, is slowly corroded by air and water.

presence of such weak acids as carbonic acid, producing a green, porous, basic carbonate copper. Green corrosion products, called verdigris or patina, appear on such copper alloy, brass and bronze, as well as on pure copper.

~~*~~ Some metals, called noble metals, are so inactive chemically that they do not suffer corrosion from the atmosphere; among them are silver, gold, and platinum. A combination of air, water, and hydrogen sulfide will act on silver, but the amount of hydrogen sulfide normally present in the atmosphere is so small that the degree of corrosion is negligible except for the black discoloration, called tarnishing, produced by the formation of silver sulfide. The corrosion of metals is more of a problem than that of other materials. Glass is corroded by strongly alkaline solutions and concrete by sulfate-bearing waters. The corrosion resistance of glass and concrete can be greatly increased by changes in their composition

4.1.2 Types of Corrosion

There are several types of corrosion, they are as follows

- I. Uniform corrosion
- II. Pitting corrosion
- III. Crevice corrosion
- IV. Microbiological corrosion
- V. Atmospheric corrosion
- VI. Cavitation corrosion
- VII. Inter-granular corrosion
- VIII. Fretting corrosion
- IX. Erosion corrosion
- X. Stress corrosion

4.1.3 Major Factors Affecting Corrosion: Some of the major factors that affect corrosion are as follows

- i. Time of exposure
- ii. Temperature
- iii. Oxygen concentration of the environment
- iv. Usage of the material or metal

V. Composition and structure of metals

4.1.4 Basic importance of Corrosion Prevention

1. Corrosion prevention: is of significance in economy and include the objectives of material losses resulting from the corrosion of pipes, tanks, metals components of machine, ships, bridges and marine structure.
 2. To improve safety of operational equipment which through corrosion may fail with catastrophic consequence e.g. pressure vessels, boilers, metallic containers for radioactive material, turbine blades and rotor.
 3. The conservation applied primarily to metal resources, the World's supply of which is limited and the wastage includes corresponding losses of energy and water reserves associated with the production and fabrication of metals structures
- #### 4.1.5 Corrosion Prevention : We have four (4) ways of corrosion prevention, viz
- a) Cathode protection
 - b) Electroplating
 - c) Painting
 - d) Galvanizing

i. Painting: ships, agricultural machines and equipment, iron bridges e.g. are painted to prevent them from useless corrosion attack

ii. Electroplating: any metal higher than iron in the electrochemical series, such as aluminum, magnesium, Zinc will be placed on top of iron rising direct current, during this process the metal would deposit itself. So that, it deposited metal will prevent iron from corrosion, since any of the plated metal does not react with either Oxygen or vapors.

iii. Galvanizing: in this methods Zinc powder is used in spraying iron and steel and thereafter, a protective layer of Cd or Zn oxides builds up over the corroded or scratched surface. Galvanic protection is possible when the surface metal is higher in the galvanic series, i.e. Cd and Zn on steel or Iron. Galvanizing, process of coating a base metal, such as iron or steel, with a thin layer of zinc to protect the base metal from corrosion. Zinc is applied with greater ease and at lower cost than other metallic coatings such as tin, chromium, nickel, or aluminum. The zinc layer protects the base

metal even when there are cracks or small gaps in the coating, because oxygen more readily with zinc than with the exposed base metal. *

The most widely used method of galvanizing is the *hot dip* process. The iron or base metal is pickled (immersed in acid) to remove dust, dirt, and grease. It is then washed and dipped into the *spelter*, that is, molten zinc. In a different galvanizing process known as *sherardizing*, the article to be galvanized is covered with zinc dust and heated in a tight closed drum for several hours at 300° to 420° C (572° to 788° F). Other methods of galvanizing include depositing the zinc electrolytic ally and applying molten zinc in the form of a fine spray. Some examples of commonly galvanized products are garbage cans, corrugated sheets for roofing, iron pipe, and fencing wire.

iv. Cathode Protection: this method is commonly used at the sea port where ships are being loaded and off-loaded. The sea water consists of Magnesium and Sodium in large amount. A piece of Magnesium is attached to the steel steps through which the ship is loaded and off-loaded and allowed it to be in contact with sea water.

The sea water acts as an electrolyte, Magnesium now acts as cathode, since it is higher in the electrochemical series, negligibly since there are Magnesium salt in the sea water and Magnesium does not react and faster with steam. Hence, iron is free from any corrosion.

4.1.6 Rusting

Rusting is a chemical process that can take place in metals exposed to the atmosphere. Not all metals however, rust. Rusting is common with metal and iron, certain conditions in the air within metal have to be present for iron to rust. Iron rust generally looks like brownish in color.

1. **Reddish brown coating on metal**: a reddish brown coating of iron oxide on the surface of iron or steel that forms when the metal is exposed to air and moisture

2. **Something resembling rust**: something that resembles rust, especially in color, or another type of corrosion or a stain

The rusting of iron can be prevented in towns and cities where iron sheet are used for roofing house. Rusting is scientifically called oxidation, which occurs when Oxygen comes in long-term contact with certain metals. Overtime, the Oxygen combine with the metal at an atomic level, forming a new compound called an Oxide and weaken the bonds of metals itself. The thinner the metal the better chance of rusting will occur, but if the metal is thick, the rusting process will take long term.

Hence, rusting is an electrochemical process which required the presence of

(i) Water (ii) Oxygen (iii) Electrolyte

In the absence of any of this, rusting does not occur to any significant extent, catalyst for the rusting process is water or di-hydrogen oxide.



Corrosion	Rusting
It is caused by chemical or electrochemical reaction	It is caused by the action of oxygen and moisture
Corrosion is the broad term for something that is being broken down due to chemical reaction	Rust is the specific type of corrosion that consists of iron being reacting with water or moist air.
Is the chemical process by which the metal waste away	Rusting is, when the metal corrode

4.2

ENGINEERING MATERIALS

4.2.1 Factors to be Consider When Selecting Engineering Materials

In order to succeed in the design process, considerable care is needed in the selection of the material that will meet the requirement of the final product. Improper choice of material will results in generation of scraps, high production time, low productivity and product unit price. Material selection is the final, practical decision in the engineering

design process can be determine that designs ultimate success or failure. There are two separate decisions to be made: firstly, one must decide which type of material that is appropriate (i.e. metals or ceramics)

Secondly, the best specific material within that category must be found e.g. Magnesium alloy preferable to aluminum or steel. The following factors are necessary for consideration before selecting materials

- (i). Manufacturing Process
- (ii) Material Property
- (iii) Compatibility
- (iv) Environmental Impact
- (v) Material Cost
- (vi) Availability

4.2.2 Mechanical Properties of Engineering Materials

Mechanical properties are those properties associated with the ability of the material to resist mechanical forces and load being applied, they include the following

- (i) *Strength*: this is the most significant mechanical property, since it is this property which governs the working stress value used in design. The strength of a material will depend upon how it is loaded, very important considering the stress analysis of a component subjected to stresses on two and three dimension.
- (ii) *Ductility*: it is the property of a material enabling it to be drawn into wire with the application of a tensile force. A ductile material must be both strong and plastic, the ductility is usually measured by the terms, percentage elongation and percentage reduction in area.
- (iii) *Stiffness*: is the ability of a material to resist deformation under stress, the modulus of elasticity is the measure of stiffness. Examples of stiff components are springs, shafts and connecting rods.
- (iv) *Elasticity*: is the property of a material to regain its original shape after deformation when the external forces are removed. This property is desirable for material used in tools and machine.

- (v) *Plasticity*: is the property of a material which enables it to retain the deformation produced under load permanently. This property of the material is necessary for forgings, in stamping images on coins and in ornamental work.
- (vi) *Brittleness*: is the property of a material opposite to ductility, it is the property of breaking of material with little or no permanent deformation. e.g. cast iron and brass, concrete is brittle when tensile load is applied to it.

(vii) *Toughness*: is the property of a material to resist fracture due to high impacts loads like hammer blows, toughness decreases when material is heated. Examples of tough material are cutting tools, car bumpers and springs

- (viii) *Malleability*: is the property of a material which enables a metal to withstand deformation by a compressive force without fracture occurring. It allows metal to be forged or rolled. Malleability of most metals increases at higher temperature and for this reason much forming of metals is done by hot rolling, forging and stamping e.g. Aluminum, Tin, Copper, Lead and Zinc.

- (ix) *Hardness*: is the property which enable a material to resist abrasion and indentation, hardness is an essential property of cutting tools materials and of components which resist wear. Hardness can be measured by Brinell, Vickers or Rockwell methods.

- (x) *Machinability*: is the ease or difficulty with which a metal can be machined
e.g. cast iron and brass are machinable material

4.2.4 Physical Properties of Engineering Material

- a) *Corrosion resistance*: The resistance against atmospheric oxygen to react with metals and form oxides. Corrosion resistance metals include Aluminum, Zinc, Tin, Chromium, Lead and Copper etc. Plastics are also corrosive resistant materials. This property is mostly used in food processing and in transportation or storage of corrosive liquids and chemicals
- b) *Fluidity*: This property is very useful in casting process, it is defined as the ability of a molten metal to flow easily and occupy every cavities parts of a mold thus enabling the casting of intricate shapes. The relatively large amount of

carbon in cast iron approximately 3 1/2 % helps this metal to become very fluid in the molten state thus allowing intricate casting to be produced. Plain carbon steel on the other hand contains much 0.1 - 1.5%. The steel therefore does not become very fluid in the molten state and intricate plain carbon steel casting cannot be produced. END OF T1 (2-7-2022)

5.1.1

STRESS

It is a measurement of density of force. It is defined as force per unit area of cross section. The SI units of stress is the Pascal (Pa) which is equivalent to a force of one Newton acting on an area of one square meter (N/m^2 or NM^{-2}).

$$\text{stress} = \frac{\text{Force (F)}}{\text{Cross Sectional Area}}$$

Worked Example:

A steel rod 6mm in diameter is under the action of a force 400N. Calculate the stress in the bar.

Solution

$$\text{stress}(\sigma) = \frac{\text{Force (F)}}{\text{Area (A)}} = \frac{400}{\pi \times (0.003)^2} \quad \pi = \frac{22}{7} =$$

$$A = \pi r^2 \text{ Or } \pi \frac{d^2}{4}$$

$$\text{Stress} = 14147072 \text{ N/m}^2$$

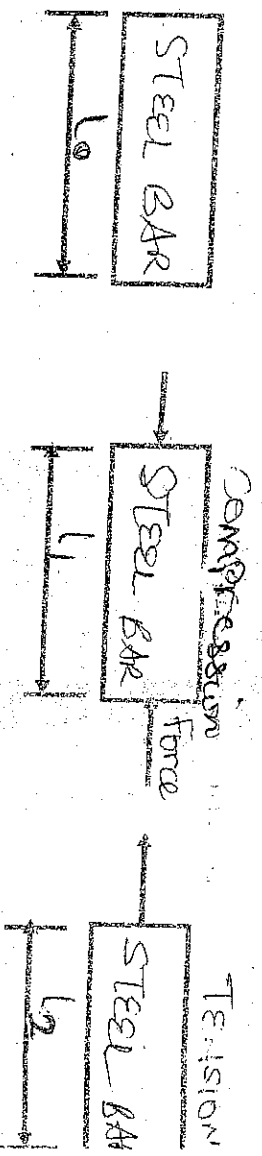
$$A = \pi \frac{d^2}{4} = \frac{3.142 \times (0.006\text{m})^2}{4}$$

$$= 2.8278 \times 10^{-5} \text{ m}^2$$

5.1.2

STRAIN

This refers to the proportional deformation produced in a material under the influence of stress; it is measured as the number of millimeter of deformation suffered per millimeter of original length and is a numerical ratio, i.e. it has no units.



$$\text{strain} = \frac{\text{increase in length}}{\text{original length}} = \frac{l - l_0}{l_0} \quad (\text{TENSION})$$

where, l_0 = original length, l = final length and l_2 = final length

$$\text{Strain} = \frac{\text{decrease in length}}{\text{original length}} = \frac{l_0 - l_1}{l_0} \quad (\text{COMPRESSIVE})$$

The original gauge length (l_0) marked on an aluminum test piece is 40mm, the test piece is strained in tension, so that the gauge length becomes 42.3mm. Calculate the strain (ϵ)

Solution:

$$\text{strain } (\epsilon) = \frac{\text{increase in length}}{\text{original length}} = \frac{l - l_0}{l_0}$$

$$= \frac{42.3 - 40}{40} = \frac{2.3}{40} = 0.0575$$

Strain (ϵ) is usually quoted as a percentage i.e.

$$\epsilon = 0.0575 \times 100\% = 5.75\%$$

5.1.3 YOUNG MODULUS OF ELASTICITY (E)

Is the ratio between the stress applied and the elastic strain it produce, that is, is the stress required to produced a unit quantity of elastic strain. The modulus of elasticity is expressed in terms of either tensile or compressive stresses and their units are the same as those for stress, since,

$$E = \frac{\text{stress}}{\text{strain}} = \frac{N/m^2}{m/m} = N/m^2 = Pa$$

Because of the numerical magnitude of the value of E, it is commonly expressed as GPa (Gin/m^2)

Worked example:

A steel wire 0.55mm^2 in cross-sectional (A) and 10m long is extended elastically 1.68mm by a force of 17.24N . Calculate the modulus of elasticity (E) for the steel.

Solution

$$\text{stress } (\sigma) = \frac{\text{force } (P)}{\text{area } (A)} = \frac{17.24}{0.55} \text{ N/mm}^2 = \frac{17.24}{0.55 \times 10^{-6}} \text{ N/m}^2 = 31.34 \text{ MN/m}^2$$

$$\text{strain } (\epsilon) = \frac{1.68 \times 10^{-3} \text{ m}}{10 \text{ m}} = 0.000168 \quad 0.55 \text{ mm}^2 = 0.55 \times 10^{-6}$$

$$\text{Young modulus } (E) = \frac{\text{Stress } (\sigma)}{\text{Strain } (\epsilon)} = \frac{31.34 \text{ MN/m}^2}{0.000168} = \frac{186547 \text{ MN}}{\text{m}^2} = 186.5 \text{ GPa} \quad 23/7/2022$$

5.2

STRESS/STRAIN DIAGRAM

When corresponding values of stress and strain derived during a tensile test are plotted graphically it is found that each type of material is represented by a characteristics curve. Material of negligible ductility such as fully hardened steels, cast iron and concrete undergo little or no plastic deformation before fracture, that is, there is no yield point and only elastic extension occurs.

A ductile material on the other hand exhibits an elastic limit (or limit of proportionality) beyond which plastic deformation occurs.

- Yield strength is the maximum stress which a material can withstand before plastic flow sets in.

6.1

MAJOR TEST FOR HARDNESS OF MATERIAL

6.1.1 HARDNESS: this is define as the ability of engineering material to resist surface abrasion. The relative hardness of minerals have in fact long been assessed by reference to Mohr's scale. These consists of a list of material arranged in order such that any minerals in the list will scratch any one below it. Thus diamond the hardest substance known, leads the list with a hardness index of 10, while talc is at least with a hardness index of 1. The surface hardness of any substance can be related to Mohr's scale by determine which of this standard will just scratch it. Mohr's scale hardness expresses numerical values for

some minerals used as standard for measuring hardness of any substance. The following table shows minerals and its hardness index.

S/NO	MINERALS	HARDNESS INDEX
1.	Diamond	10
2.	Corundum	9
3.	Topaz	8
4.	Quartz	7
5.	Orthoclase feldspar	6
6.	Apatite	5
7.	Fluorite	4
8.	Calcite	3
9.	Gypsum	2
10.	Talc	1

6.1.2 Major Test for Hardness of a Material

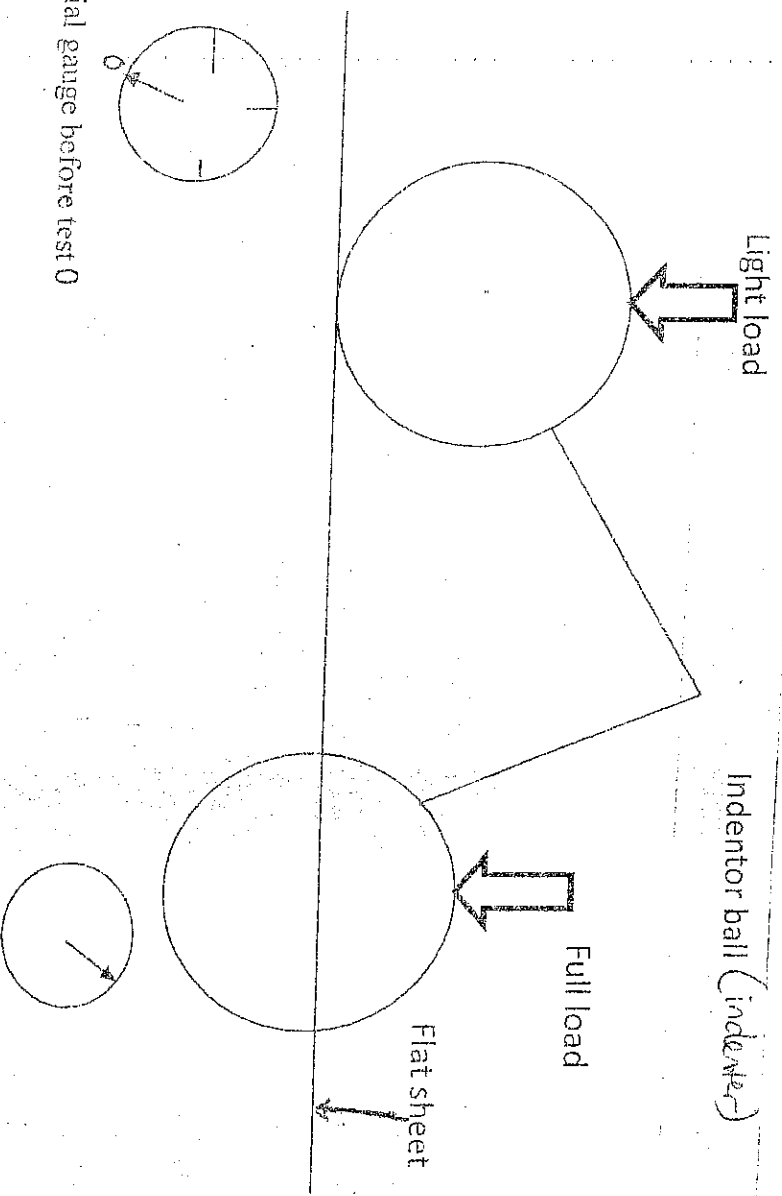
The following are major tests for hardness of engineering materials

- i. Rockwell hardness test
- ii. Brinell hardness test
- iii. Knoop and Vickers micro indentation hardness

6.1.3 Rockwell Test: The depth of indentation is measured directly and thus avoiding the use of microscope, this eliminates the time consuming operation and the test are quicker than Brinell or Vickers tests. Rockwell hardness machines are frequently used on the shop floor for quality control checks. The machine is fitted with a dial gauge which indicates the depth to which the indenter has penetrated. The sequence of operation are as follows

- i. A light load is applied to the indenter to ensure a good contact with the metal surface.

- ii. Full load is applied forcing the indenter into the metal.
- iii. Full load is removed leaving a light load to maintain contact
- iv. The dial now indicates the depth of indenter(D)



Dial gauge after test

6.1.4 The Brinell Test: It was invented by a Swedish, Dr Johann Brinell in 1900, in this test a hardened steel ball is force into the surface of the test piece using an appropriate specified force, the diameter of the impression produced is measured and the Brinell hardness index, H_B derived from

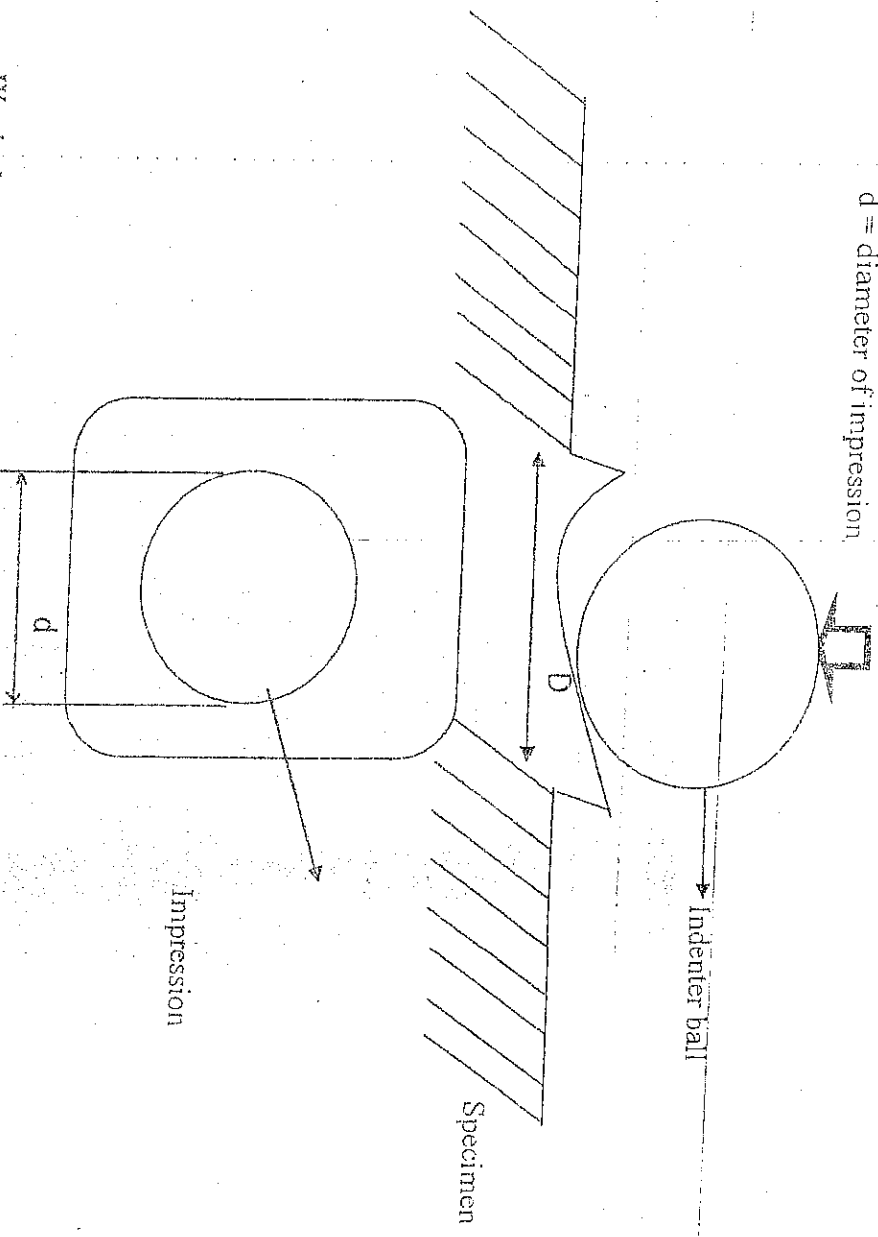
$$\text{Brinell hardness } (H_B) = \frac{\text{Force } (P)}{\text{Surface area of impression}}$$

The surface area of the impression can be shown to be

$$\frac{\pi D}{2} \left(D - \sqrt{D^2 - d^2} \right)$$

Where D = diameter of the indenter ball

d = diameter of impression



Worked example:

A 25mm diameter hardened steel ball was used as an indenter and a force of 100N was applied, the diameter of the impression of ~~the~~ ^{measured} ~~measured~~ ^{measured} with a calibrated microscope is given as 15mm, calculate the Brinell hardness number

Solution:

$D = 25\text{mm}$ = diameter of the indenter ball given

$d = 15\text{mm}$ = diameter of the impression given

$$\begin{aligned}
 \text{Surface area of impression} &= \frac{\pi D}{2} (D - \sqrt{D^2 - d^2}) \\
 &= \frac{3.142 \times 0.025}{2} (0.025 - 0.02) \\
 &= \frac{3.142 \times 0.025}{2} (0.005) \\
 &= 0.039275 \times (0.005) \\
 &= 1.96375 \times 10^{-4}
 \end{aligned}$$

Surface area of impression $\approx 1.964 \times 10^{-4} \text{m}^2$

$$\begin{aligned}
 H_b &= \frac{\text{Force (P)}}{\text{Surface area impression}} \\
 &= \frac{100}{1.964 \times 10^{-4}} = 509,165 \text{N/m}^2 \\
 &= 509.17 \text{KN/m}^2 \quad 5/11/1022
 \end{aligned}$$

6.15 The Vickers Diamond Pyramid Test (or Diamond Pyramid Hardness Test)

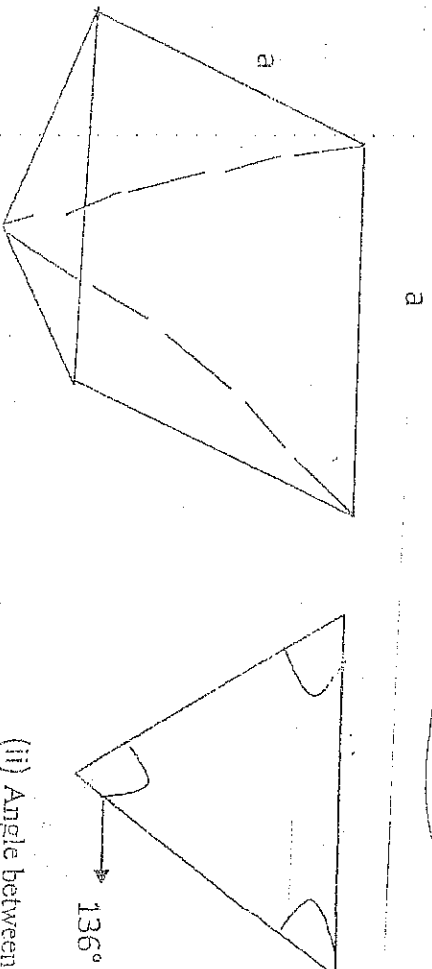
It uses a diamond pyramid indenter of geometrical shape indicated in next page. Vickers test has the following advantages over Rockwell and Brinell test which use spherical steel indenter.

- i. The pyramid is of constant angle making the choice of a suitable $\frac{P}{d^2}$ ratio unnecessary.
- ii. The diamond pyramid is less likely to distort under high forces than a steel ball. The Vickers hardness (H_v) is related to the applied force (P) and the surface area of surface area of impression in the same manner as in Brinell hardness

$$\begin{aligned}
 H_v &= \frac{\text{force (P)}}{\text{Surface area of impression}} \\
 &= \frac{2P \sin \frac{\theta}{2}}{d^2}
 \end{aligned}$$

Where $\theta = 136^\circ$ between opposite faces in the diamond pyramid and d is the mean length of diagonals (d_1 and d_2) of the impression in (mm). Hence $= \frac{2p \sin \frac{136^\circ}{2}}{d^2}$

$$H_v = \frac{2p \sin 68^\circ}{d^2} = \frac{1.8544P}{d^2} \text{ N/m}^2$$



(ii) Angle between opposite faces of

(i) Diamond indenter

Used in Vickers test

$d = \frac{1}{2} (d_1 + d_2)$ = the impression produced in the test piece

Worked example:

Calculate the Vickers hardness of a material when a force of 25N is applied to the pyramid shaped indenter whose average length is 2.5mm

Solution

$$H_v = \frac{1.8544P}{d^2} \text{ where } P \text{ is the force on the material and } d$$

$= \text{average length}$

$$H_p = \frac{1.8544 \times 25}{(0.0025)^2} = \frac{46.36}{6.25 \times 10^{-6}}$$

$$H_p = 7417600 \text{ N/m}^2 = 7417.6 \text{ kN/m}^2$$

$$H_p = 7.42 \text{ MN/m}^2$$

7.1

METALLIC STRUCTURE

The crystal structures in metal are forces binding, the atoms in a metal which are non-directional. All atoms of pure metal are of the same kind and size arrange themselves in the closest possible packing patterns which are associated with position of minimum potential energy. Most of the metals of engineering, importance crystallize in one of three principal types of structure in all of which the atoms are held together solely by the metallic bond. The most crystal structures or metallic structures are as follows

- i. Body centered cubic (BCC)
- ii. Face centered cubic (FCC)
- iii. Close Packed Hexagonal (CPH)

7.1.1 Body Centered Cubic (BCC)

In this type of structure an atom occupies each corner of an imaginary cube with an additional atom at the center of the cube. BCC metals are less ductile and ductility exists over a relatively small temperature range. Example of metals with BCC are α -iron, Tungsten, Molybdenum and Vanadium

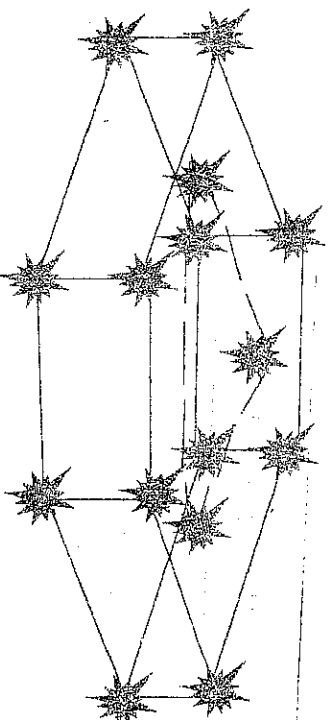
7.1.2 Face Centered Cubic (FCC)

In this type of structure an atom occupies each corner of an imaginary cube with an additional atom at the center of each face of the cubic. FCC metals are nearly and always ductile over a wide range of temperature. Examples of metals having FCC are Platinum, Copper, Aluminum, Lead.

7.1.3 Closed Packed Hexagonal (CPH)

In this type of structure, atom occupies each corner of an imaginary right prism with an atom at the centered of the top and bottom of the hexagon and three atoms equally spaced in the centered of the prism. CPH metals have low ductility which exists over a small temperature outside this temperature range, they are brittle. Examples includes; α -iron, Zinc, Magnesium, Cadonitium and Beryllium

CPH



6/8/2022

8.1 FERROUS AND NON FERROUS METALS: Metals are classified as ferrous and non-ferrous metals

8.1.1 Ferrous Metals

Ferrous metals are iron as the major constituent such as steel, cast iron, mild steel wrought iron and iron- carbon steel. They are chosen for strength with cheapness, wide availability and general convenience of forming or machining for which the steels especially can be given a wide range of properties by suitable alloying. The most commonly used in industry are wrought iron, cast iron and iron-carbon alloy steel. These three materials are defined mainly by percentage of carbon which they contain, wrought iron is almost pure iron free from carbon, steel may contain from 0.05 % - 2 % carbon.

Wrought iron is used mainly for chain-links coupling, hooks and cranes, while it is good ductility and reduces the possibility of fracture, cast iron in general is cheap to produce, easy to cast and it can machine well. Mild steel contains 0.05 % - 0.3 % carbon, medium carbon steels contains from 0.3 % - 0.5 % carbon, while high carbon steel contains from above 0.5 % up to 2 %.

8.1.2 Non-Ferrous Metals

These are metals that do not contain iron. They are usually more costly than ferrous metals. Characteristic of non-ferrous are as follow

- i. Good thermal conductivity
- ii. Non – magnetic properties
- iii. Good machinability

Example of non-ferrous metal are as follow: Copper (Cu), Aluminium and its alloys, alloys the combination of Copper and Zinc, Tin, Lead and Zinc

8.2

PHYSICAL PROPERTIES AND USES OF NON-FERROUS METAL

8.2.1 Physical Properties of Copper (Cu) and its alloy

- i. Copper has a very high specific conductivity
- ii. High melting point of 1083°C
- iii. They are malleable
- iv. Posses good conductivity of heat and electricity
- v. There are ductile
- vi. Good corrosion resistance

8.2.2 Uses of Copper and its Alloy

- i. Mainly used in the electrical industries
- ii. Copper and its alloy are used to resist corrosion

8.2.3 Aluminium and its alloy

- i. It is ductile, malleable, light and it is good conductor of electricity and heat
- ii. Aluminium alloys is very strong in strength
- iii. It has a melting point of 660°C
- iv. Pure Aluminium is weak i.e. little strength

8.2.4 Uses of Aluminium

- i. Aluminium alloys are used in foundry workshop for casting, rolling and forging
- ii. Aluminium are used in roofing Zinc and so many others house hold utensil

8.2.5 Physical Properties of Lead (Pb)

- i. Melting point of 327°C
- ii. Leads form its alloy with Tin and Copper
- iii. Good resistance to corrosion

Uses: it is used in bearing of metals and machine

8.2.6 Physical Properties of Tin and its alloy

- i. Melting point of 232°C
- ii. It is harder than Lead.
- iii. It has good resistance to corrosion
- iv. The combination of copper and tin or aluminium is known as bronze
- v. Very expensive
- vi. Possesses the property of ductility
- vii. Very weak in strength

Uses: (i) When form an alloys with copper, it is used in the formation of bronze

- (ii) It is used in resistance to corrosion

9.1 POLYMER (PLASTIC)

A natural or synthetic compound that consist of a large molecules made of many chemically bonded smaller identical molecules e.g. starch and Nylon, this molecular structure from organic molecules is known as **Polymerizations**. Polymer includes rubber, plastics and many types of adhesives. polymer have low electrical and thermal conductivities are low in strength and are not suitable for use at high temperature.

9.1.1 Plastics: They are extremely versatile synthetic material made from the polymerization of organic compound. It can be molded into various shapes or fabricated in many different forms for use in commerce and industry. What is as hard as stone, strong as steel; clear as glass, light as wood, and springy as rubber? Plastics can be like all of these things. Plastics are materials used to make a wide variety of products. Plastics cost less than many other materials. Plastics do not rot like wood. They do not rust like metals. Plastics can be any color. You use plastics every day. Your computer case is made of hard, strong plastic. You drink out of cups made of light plastic foam. You wear clothes and walk on carpets made of plastic threads. You buy food and other goods covered in clear plastic wrap. Plastic is everywhere around you.

They are highly versatile and useful material also higher chemical reactivity.

Classification of polymer: these are classified as follow

- i. Thermoplastic polymer: occurs in which the long molecule chains are not rigidity connected have good formability
- ii. Thermosetting plastics: are much stronger but more brittle because the molecular chains are rigidly linked. Polymer is used in much application which includes electronic devices. Example of polymers is as follows:
 - Polyethylene: application in food package they are easily formed into thin, flexible, airtight film.
 - Phenolic: used as adhesives for joining plywood, plastics as well as metal, they are strong and have good moisture-resistance. A time phenolic is a resin that has high temperature stability. Use: In plastics, paints and adhesives.
 - Epoxy: used for encapsulating integrated circuit in the electronic industries, they possess good electrical insulation and moisture resistance. Epoxy is to stick one thing to another using epoxy resin

9.1.2 Physical properties of Polymer (Thermosetting)

- i. Have low electrical and thermal conductivities
- ii. Low in strength

- iii. Not suitable for use at high temperature
- iv. Have high chemical reactivity
- v. Highly versatile and useful materials

9.1.3 Where Does Plastic Come From?

Plastic comes mainly from *fossil fuels*. Oil, coal, and natural gas are fossil fuels. The building blocks of fossil fuels are the chemicals hydrogen and carbon. Like all chemicals, hydrogen and carbon are made from *atoms*, tiny bits of matter much too small to see. Everything is made of atoms. You make plastic by heating fossil fuels and adding other chemicals. The hydrogen and carbon atoms link together. The atoms join together like beads on a string. The long chains of atoms are what make plastics special. Metal, glass, wood, and clay are made of short chains of atoms.

9.1.4 Molding Plastic

Plastic starts out as hot, sticky goo, called *resin*. The resin gets made into car parts, compact disks, plastic bottles, and all the different kinds of plastic products.

To make plastic products, the heated resin is poured or forced into a mold. The resin cools and hardens into the shape of the mold. In fact, the word *plastic* comes from Latin and Greek words that mean *to mold*.

9.1.5 Does Plastic Harm The Environment?

People in the Nigerians throw millions of tons of plastic away every year. Plastics do not easily break down, or *degrade*. Plastics fill up garbage dumps, called *landfills*. Plastics are a major source of pollution. Using less plastic helps reduce the amount of plastic waste. Making thinner plastic bottles and other containers uses less plastic.

Many plastics can be *recycled*, or reused. Recycled plastics are used to make new plastic

11.1

HEAT: How is heat like a river? Answer: They both flow "downhill."

You will never see a river flow up a mountain. A river only goes one way, down. Heat also goes one way. Heat only flows from something that is hotter to something that is colder. You might think that putting an ice cube in your lemonade makes the lemonade cooler. What really happens is that the lemonade makes the ice cube warmer. Heat goes out of the warmer lemonade into the colder ice cube. Heat only flows one way.

11.1.1 What is Heat? Heat is an important kind of energy. Heat from the Sun makes life on Earth possible. Heat makes your home warm. Heat cooks your food. Heat makes cars go. You use more heat than any other kind of energy. Things that make heat are called heat sources. The Sun is a heat source. Fire is a heat source. Melted rock and metal inside Earth are sources of heat. Heat sources make other things hot.

11.1.2 Why do Things Get Hot? Things get hot because tiny bits of matter called atoms and molecules move around. Everything is made of atoms or molecules. They are much too small for you to see. Atoms and molecules are always moving around. They move around slowly in something that is cold. They move around quickly in something that is hot. Heat energy comes from moving atoms and molecules. An iron frying pan is made of iron atoms. Heat energy from a fire makes the iron atoms move faster. The atoms bang into one another. The faster the atoms whiz around, the hotter the frying pan gets. The hot frying pan in turn makes things like strips of bacon hot.

11.1.3 How does Heat Move? Heat energy can move in three ways. These three ways are called conduction, convection, and radiation. Conduction carries heat through things that are

solid. The frying pan heats the bacon through conduction. The frying pan sits on a flame. Heat from the flame makes atoms at the bottom of the pan move faster. The atoms on the bottom bang into the atoms above them. This makes those atoms move faster and bang into more atoms. As atoms bang into other atoms, the heat energy travels from the bottom of the frying pan to the top. The heat energy enters the bacon and cooks it to a crisp.

Convection heats liquids and gases. A radiator heats air in a room by convection. Air near the radiator gets hot. Hot air rises. Cooler air replaces the heated air. This air gets hot and also rises. The air keeps going around and around from cooler to hotter. The way the air moves is called convection current. Radiation can send heat across empty space. This is how heat from the Sun travels to Earth. Heat rays from the Sun strike the surface of Earth and make it warm. ~~the~~ ^A ~~sun~~ ^{way}.

11.1.4 TEMPERATURE: IT IS THE DEGREE OF HOTNESS / COLDNESS IN A SUBSTANCE. ITS UNIT IS MEASURED IN °C(CELSIUS) / °F(FAHRENHEIT)

CONVERSION FROM CELSIUS TO FAHRENHEIT: $9^{\circ}\text{C} = 5(^{\circ}\text{F} - 32)$

EXAMPLE: CONVERT 212°F TO $^{\circ}\text{C}$

$$^{\circ}\text{C} = [5(212 - 32)] / 9$$

$$= [180 \times 5] / 9$$

$$= 100$$

THEREFORE,

$$212^{\circ}\text{F} = 100^{\circ}\text{C}$$